



**SILVER OAK  
UNIVERSITY**  
EDUCATION TO INNOVATION

|                     |                                   |                       |            |
|---------------------|-----------------------------------|-----------------------|------------|
| <b>Semester:</b>    | 5th                               | <b>Academic Year:</b> | 2024-25    |
| <b>Course Name:</b> | Analysis and Design of Algorithms | <b>Course Code:</b>   | 1010043316 |

## Question Bank

### UNIT-1 Basics of Algorithms and Mathematics

| SR NO. | Question Text                                                                                                                                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | What is an algorithm? What do you mean by correct algorithm? What do you mean by instance of a problem? List out the criteria that all algorithms must satisfy. On what bases will you consider algorithm A is better than algorithm B? |
| 2      | What is an Algorithm? What do you mean by linear inequalities and linear equations? Explain asymptotic notation with the help of example.                                                                                               |
| 3      | Why do we use asymptotic notations in the study of Algorithms? Briefly describe the commonly used asymptotic Notations.                                                                                                                 |
| 4      | Explain master theorem and solve the following recurrence equation with master method<br>1. $T(n) = 9T(n/3) + n$<br>2. $T(n) = 3T(n/4) + n \lg n$<br>3. $T(n) = T(2n/3) + 1$                                                            |
| 5      | What is an amortized analysis? Explain accounting method and aggregate analysis with suitable example.                                                                                                                                  |
| 6      | Explain following terms with example.<br>1. Set 2. Relation 3. Function                                                                                                                                                                 |
| 7      | Do as directed.<br>Calculate computation time for the statement t3 in following code fragment?<br>for i = 1 to n<br>{<br>for j = 1 to i<br>{<br>c = c + 1 ..... t3<br>}<br>}<br>2. Prove that $T(n) = 1+2+3+\dots+n = \Theta(n^2)$ .    |
| 8      | Define an amortized analysis. Briefly explain its different Techniques. Carry out aggregate analysis for the problem of implementing a k-bit binary counter that counts upward from 0.                                                  |

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|----|--------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Define following terms<br>(i) Quantifier<br>(ii) Algorithm<br>(iii) Big 'Oh' Notation<br>(iv) Big 'Omega' Notation<br>(v) 'Theta' Notation |
| 10 | SHORT QUESTIONS<br>(i) what is an Algorithm?<br>(ii) what is worst case time complexity?<br>(iii) Big Oh notation                          |
| 11 | Define Algorithm, Time Complexity and Space Complexity.                                                                                    |
| 12 | Solve the recurrence $T(n) = 7T(n/2) + n^3$                                                                                                |

### **UNIT-1 Analysis of Algorithm**

| SR NO. | Question Text                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------|
| 1      | What is an algorithm? Explain various properties of an algorithm.                                                    |
| 2      | What do you mean by asymptotic notations? Explain.                                                                   |
| 3      | Write a program/algorithm of selection sort methods. What is complexity of the method?                               |
| 4      | Explain different asymptotic notations in brief.                                                                     |
| 5      | What is an amortized analysis? Explain aggregate method of amortized analysis using simple example.                  |
| 6      | Explain why analysis of algorithm is important? Explain: Worst case, Best case, Average case complexity.             |
| 7      | Define: Big Oh, Omega and Big Theta notation.                                                                        |
| 8      | What is Recursion? Give the implementation of Tower of Hanoi Problem using recursion.                                |
| 9      | Explain why analysis of algorithm is important?                                                                      |
| 10     | Explain bubble sort algorithm. Derive the algorithmic complexity in best case, worst case and average case analysis. |
| 12     | Explain the heap sort in detail. Give its complexity.                                                                |
| 13     | Sort the letters of word "DESIGN" in alphabetical order using bubble sort.                                           |
| 14     | Write an algorithm for insertion sort. Analyze insertion sort algorithm for best case and worst case.                |
| 15     | Explain Counting sort with example.                                                                                  |
| 16     | Sort the following data with Heap Sort Method: 20, 50, 30, 75, 90, 60, 25, 10, and 40. And explain it.               |
| 17     | What is an amortized analysis? Explain aggregate method of amortized analysis using suitable example.                |
| 18     | Explain Selection Sort Algorithm and give its best case, worst case and average case complexity with example.        |
| 19     | Sort the letters of word "EDUCATION" in alphabetical order using insertion sort.                                     |

|    |                                                                                                                                                                              |
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| 20 | Apply the bubble sort algorithm for sorting {U,N,I,V,E,R,S}                                                                                                                  |
| 21 | Let $f(n)$ and $g(n)$ be asymptotically nonnegative functions.<br>Using the basic definition of $\Theta$ -notation, prove that<br>$\max(f(n), g(n)) = \Theta(f(n) + g(n))$ . |
| 22 | What is the smallest value of $n$ such that an algorithm whose running time is $100n^2$ runs faster than an algorithm whose running time is $2^n$ on the same machine?       |
| 23 | Explain Tower of Hanoi Problem, Derive its recursion equation and computer it's time complexity.                                                                             |

## UNIT-2 Divide and Conquer Algorithm

| SR NO. | Question Text                                                                                                                                                                                                                                                                                                                     |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Prove that Greedy Algorithms does not always give optimal solution. What are the general characteristics of Greedy Algorithms? Also compare Greedy Algorithms with Dynamic Programming and Divide and Conquer methods to find out major difference between them.                                                                  |
| 2      | Justify the general statement that “if a problem can be split using Divide and Conquer strategy in almost equal portions at each stage, then it is a good candidate for recursive implementation, but if it cannot be easily be so divided in equal portions, then it better be implemented iteratively”. Explain with an example |
| 3      | Write an algorithm for binary search. Calculate the time complexity for each case.                                                                                                                                                                                                                                                |
| 4      | Write an algorithm for merge sort with divide and conquer strategy. Analyze each case. List best case worst case and average case complexity                                                                                                                                                                                      |
| 5      | Write an algorithm for quick sort with divide and conquer strategy. Analyze each case. In which case it performs similar to selection sort?                                                                                                                                                                                       |
| 6      | Differentiate divide and conquer with dynamic programming. Write recurrence for calculation for binomial coefficient.                                                                                                                                                                                                             |
| 7      | Explain how to apply the divide and conquer strategy for sorting the elements using merge sort.                                                                                                                                                                                                                                   |
| 8      | Differentiate the following:<br>1. Divide and conquer & Dynamic Programming<br>2. Greedy Algorithm & Dynamic Programming                                                                                                                                                                                                          |
| 9      | Show how divide and conquer technique is used to compute product of two n digit no with example.                                                                                                                                                                                                                                  |
| 10     | Sort the following list using quick sort algorithm:<br><50, 40, 20, 60, 80, 100, 45, 70, 105, 30, 90, 75><br>Also discuss worst and best case of quick sort algorithm.                                                                                                                                                            |
| 11     | Explain Binary search algorithm with divide and conquer strategy and use the recurrence tree to show that the solution to the binary search recurrence $T(n) = T(n/2) + \Theta(1)$ is $T(n) = \Theta(\lg n)$ .                                                                                                                    |
| 12     | Explain how to apply the divide and conquer strategy for sorting the elements using quick sort with example. Write algorithm for quick sort method.                                                                                                                                                                               |
| 13     | Discuss matrix multiplication problem using divide and Conquer technique.                                                                                                                                                                                                                                                         |
| 14     | Explain Strassen's algorithm for matrix multiplication.                                                                                                                                                                                                                                                                           |
| 15     | Explain the use of Divide and Conquer Technique for Binary Search Method. What is the complexity of Binary Search Method? Explain it with example.                                                                                                                                                                                |
| 16     | Write a program/algorithm of Quick Sort Method and analyze it with example.                                                                                                                                                                                                                                                       |
| 17     | Write an algorithm for quick sort and derive best case, worst case using divide and conquer technique also trace given data (3,1,4,5,9,2,6,5)                                                                                                                                                                                     |
| 18     | Multiply 981 by 1234 by divide and conquer method.                                                                                                                                                                                                                                                                                |
| 19     | What do you mean by Divide & Conquer approach? List advantages and disadvantages of it                                                                                                                                                                                                                                            |

|    |                                                                                                                                                         |
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| 20 | Solve the following recurrence relation using iteration method. $T(n) = 8T(n/2) + n^2$ . Here $T(1) = 1$ .                                              |
| 21 | Write Merge sort algorithm and compute its worst case and best-case time complexity. Sort the List G,U,J,A,R,A,T in alphabetical order using merge sort |
| 22 | Demonstrate Binary Search method to search Key = 14, form the array $A = \langle 2, 4, 7, 8, 10, 13, 14, 60 \rangle$                                    |
| 23 | Write an algorithm for insertion sort. Analyze insertion sort algorithm for best case and worst case.                                                   |

### UNIT-3 Dynamic Programming

| SR NO. | Question Text                                                                                                                                                                                                                                                                                             |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Given two sequences of characters, $P=\langle ABCDABE \rangle$ , $Q=\langle CABE \rangle$ Obtain the longest common subsequence                                                                                                                                                                           |
| 2      | Given the four matrix find out optimal sequence for multiplication $D=\langle 15,5,10,20,25 \rangle$                                                                                                                                                                                                      |
| 3      | Given coins of denominations 1, 3 and 4 with amount to be pay is 7. Find optimal no. of coins and sequence of coins used to pay given amount using dynamic method                                                                                                                                         |
| 4      | Solve following knapsack problem using dynamic programming algorithm with given capacity $W=5$ , Weight and Value are as follows: (2,12),(1,10),(3,20),(2,15).                                                                                                                                            |
| 5      | Given two sequences of characters, $P=\langle XYZYTX \rangle$ $Q=\langle YTZX \rangle$ Obtain the longest common subsequence                                                                                                                                                                              |
| 6      | Describe an assembly line scheduling problem and give dynamic programming algorithm to solve it.                                                                                                                                                                                                          |
| 7      | Using algorithm find an optimal parenthesization of a matrix chain product whose sequence of dimension is (13,5,89,3,34) (use dynamic programming).                                                                                                                                                       |
| 8      | What are the differences between greedy approach and dynamic programming?                                                                                                                                                                                                                                 |
| 9      | Generate equation for Matrix chain multiplication using Dynamic programming. Find out minimum number of multiplications required for multiplying:<br>$A[1 \times 5]$ , $B[5 \times 4]$ , $C[4 \times 3]$ , $D[3 \times 2]$ , and $E[2 \times 1]$ .<br>Also give the optimal parenthesization of matrices. |
| 10     | Discuss and derive an equation for solving the 0/1 Knapsack problem using dynamic programming method. Design and analyze the algorithm for the same.                                                                                                                                                      |
| 11     | Discuss and derive the optimal substructure that can be used to solve the Longest Common Subsequence problem using dynamic programming. Find the longest common subsequence for the given two sequences of characters:<br>$P = (1,0,0,1,0,1,1,0,1,1,0,1)$ ; $Q = (0,1,1,0)$ .                             |
| 12     | Explain principle of optimality.                                                                                                                                                                                                                                                                          |
| 13     | What is principle of optimality? Explain its use in dynamic programming method.                                                                                                                                                                                                                           |
| 14     | Discuss Assembly Line Scheduling problem using dynamic programming with example.                                                                                                                                                                                                                          |
| 15     | Explain Chained Matrix Multiplication with example.                                                                                                                                                                                                                                                       |
| 16     | Write equation for Chained matrix multiplication using Dynamic programming. Find out optimal sequence for multiplication:<br>$A_1 [5 \times 4]$ , $A_2 [4 \times 6]$ , $A_3 [6 \times 2]$ , and $A_4 [2 \times 7]$ . Also give the optimal parenthesization of matrices.                                  |
| 17     | Explain how to find out Longest Common Subsequence of two strings using Dynamic Programming method. Find any one Longest Common Subsequence of given two strings using Dynamic Programming.<br>$X=abbacdcb$<br>$Y=bcdbbcaac$                                                                              |
| 18     | Solve Making Change problem using Dynamic Programming. (Denominations: $d_1=1$ , $d_2=4$ , $d_3=6$ ). Give your answer for making change of Rs. 9.                                                                                                                                                        |
| 19     |                                                                                                                                                                                                                                                                                                           |

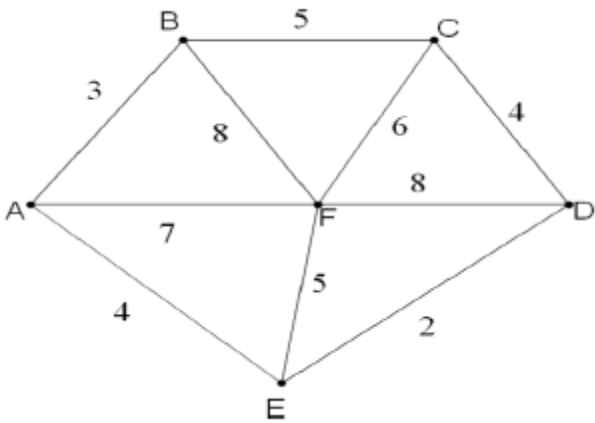
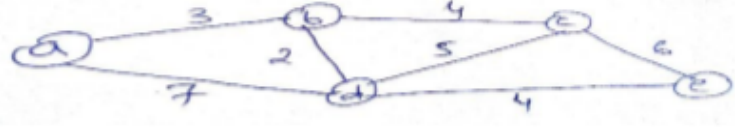
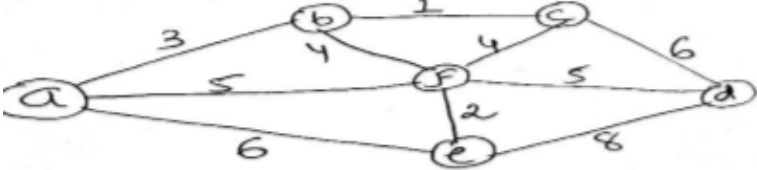
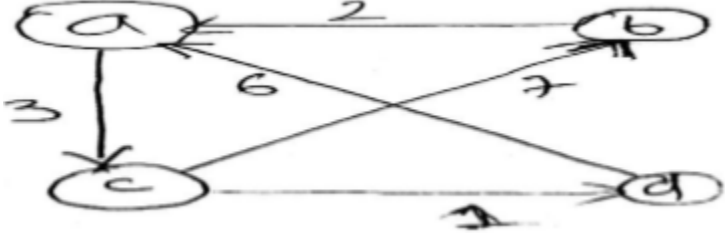
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|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Solve Making change problem using dynamic technique. $D1 = 1, d2=3, d3=5, d4=6$ . Calculate for making change of Rs. 8.                                                                                                                               |
| 20 | Given two sequence of characters, $X=\{G,U,J,A,R,A,T\}$ , $Y = \{J,R,A,T\}$ obtain the longest common subsequence.                                                                                                                                    |
| 21 | Consider Kanpsack capacity $W=50$ , $w=(10,20,40)$ and $v=(60,80,100)$ find the maximum profit using greedy approach.                                                                                                                                 |
| 22 | For the following chain of matrices find the order of parenthesization for the optimal chain multiplication (15,5,10,20,25)                                                                                                                           |
| 23 | Find optimal sequence of multiplication using dynamic programming of following matrices: $A1[10 \times 100]$ , $A2[100 \times 5]$ , $A3[5 \times 50]$ and $A4[50 \times 1]$ . List optimal number of multiplication and parenthesization of matrices. |
| 24 | Consider Knapsack capacity $W=9$ , $w = (3,4,5,7)$ and $v=(12,40,25,42)$ find the maximum profit using dynamic method                                                                                                                                 |
| 25 | Solve Making change problem using dynamic technique. $d1 = 1, d2=2, d3=4, d4=6$ , Calculate for making change of Rs. 10                                                                                                                               |
| 26 | Find out the NCR (5 3) Using dynamic method.                                                                                                                                                                                                          |

### **UNIT-3 Greedy Algorithm**

| <b>SR NO.</b> | <b>Question Text</b>                                                                                                 |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| 1             | Give and explain Kruskal's Algorithm for Minimum Spanning Tree and Compare it with Prim's algorithm with an example. |
| 2             | Is Selection sorting a greedy algorithm? If so, what are the functions involved.                                     |
| 3             | Write down the general characteristics of greedy algorithm                                                           |
| 4             | Explain Kruskal's algorithm to find minimum spanning tree with an example. What is it's time complexity?             |
| 5             | What do you mean by minimum spanning tree? Explain single source shortest path with the help of example              |
| 6             | Give and explain Prim's Algorithm for Minimum Spanning Tree and Compare it with Kruskal's algorithm with an example. |
| 7             | Explain Greedy method in detail with example and differentiate it with dynamic method.                               |
| 8             | Find Minimum Spanning Tree for the given graph using Prim's Algo. (Initialization from node A).                      |

| 9           | Explain Dijkstra's shortest path algorithm with example. If we want to display intermediate node than what change we should make in the algorithm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |                |                |                |                |                |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------|---|---|---|---|---|---|---|-------------|---|---|---|----|----|---|---|
| 10          | Mention applications of minimum spanning tree. Generate minimum spanning tree from the following graph using Prim's algorithm. (Start at vertex a).<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |                |                |                |                |                |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |
| 11          | <p>Following are the details of various jobs to be scheduled on multiple processors such that no two processes execute at the same on the same processor.</p> <table border="1"><thead><tr><th>Jobs</th><th>J<sub>1</sub></th><th>J<sub>2</sub></th><th>J<sub>3</sub></th><th>J<sub>4</sub></th><th>J<sub>5</sub></th><th>J<sub>6</sub></th><th>J<sub>7</sub></th></tr></thead><tbody><tr><td>Start time</td><td>0</td><td>3</td><td>4</td><td>9</td><td>7</td><td>1</td><td>6</td></tr><tr><td>Finish time</td><td>2</td><td>7</td><td>7</td><td>11</td><td>10</td><td>5</td><td>8</td></tr></tbody></table> <p>Show schedule of these jobs on minimum number of processors using greedy approach. Derive an algorithm for the same. What is the time complexity of this algorithm?</p> | Jobs           | J <sub>1</sub> | J <sub>2</sub> | J <sub>3</sub> | J <sub>4</sub> | J <sub>5</sub> | J <sub>6</sub> | J <sub>7</sub> | Start time | 0 | 3 | 4 | 9 | 7 | 1 | 6 | Finish time | 2 | 7 | 7 | 11 | 10 | 5 | 8 |
| Jobs        | J <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | J <sub>2</sub> | J <sub>3</sub> | J <sub>4</sub> | J <sub>5</sub> | J <sub>6</sub> | J <sub>7</sub> |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |
| Start time  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3              | 4              | 9              | 7              | 1              | 6              |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |
| Finish time | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7              | 7              | 11             | 10             | 5              | 8              |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |
| 12          | Define MST. Explain Kruskal's algorithm with example for construction of MST.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |                |                |                |                |                |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |
| 13          | Explain in brief characteristics of greedy algorithms. Compare Greedy Method with Dynamic Programming Method.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |                |                |                |                |                |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |
| 14          | Write the Prim's Algorithm to find out Minimum Spanning Tree. Apply the same and find MST for the graph given below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |                |                |                |                |                |                |                |            |   |   |   |   |   |   |   |             |   |   |   |    |    |   |   |

|           |                                                                                                                                                                                                                                                                                                                                |         |    |   |   |   |   |           |    |    |    |   |   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----|---|---|---|---|-----------|----|----|----|---|---|
|           | <pre>graph TD     0((0)) --- 3  1((1))     0 --- 6  3((3))     0 --- 1  2((2))     1 --- 5  2     3 --- 5  2     2 --- 6  4((4))     2 --- 4  5((5))     4 --- 6  5</pre>                                                                                                                                                      |         |    |   |   |   |   |           |    |    |    |   |   |
| 15        | What is recurrence? Solve recurrence equation $T(n) = T(n-1) + n$ using forward substitution and backward substitution method.                                                                                                                                                                                                 |         |    |   |   |   |   |           |    |    |    |   |   |
| 16        | Using greedy algorithm find an optimal schedule for following jobs with $n=6$ .<br>Profits: $(P_1, P_2, P_3, P_4, P_5, P_6) = (20, 15, 10, 7, 5, 3)$<br>Deadline: $(d_1, d_2, d_3, d_4, d_5, d_6) = (3, 1, 1, 3, 1, 3)$                                                                                                        |         |    |   |   |   |   |           |    |    |    |   |   |
| 17        | Write Huffman code algorithm and Generate Huffman code for following<br><table border="1"><tr><td>Letters</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>Frequency</td><td>24</td><td>12</td><td>10</td><td>8</td><td>8</td></tr></table>                                                                  | Letters | A  | B | C | D | E | Frequency | 24 | 12 | 10 | 8 | 8 |
| Letters   | A                                                                                                                                                                                                                                                                                                                              | B       | C  | D | E |   |   |           |    |    |    |   |   |
| Frequency | 24                                                                                                                                                                                                                                                                                                                             | 12      | 10 | 8 | 8 |   |   |           |    |    |    |   |   |
| 18        | <pre>graph TD     1((1)) --- 6  2((2))     1 --- 5  4((4))     1 --- 1  3((3))     2 --- 5  3     4 --- 5  3     3 --- 8  5((5))     3 --- 4  6((6))     5 --- 4  6</pre> <p>Compute MST using PRIM's Algorithm.</p>                                                                                                           |         |    |   |   |   |   |           |    |    |    |   |   |
| 19        | Find an optimal Huffman code for the following set of frequency. a : 50, b: 20, c: 15, d: 30.                                                                                                                                                                                                                                  |         |    |   |   |   |   |           |    |    |    |   |   |
| 20        | Solve the following recurrence relation using substitution method. $T(n) = 2T(n/2) + n$ . Here $T(1) = 1$ .                                                                                                                                                                                                                    |         |    |   |   |   |   |           |    |    |    |   |   |
| 21        | Suppose that we have a set of activities to schedule among a large number of lecture halls, where any activity can take place in any lecture hall. We wish to schedule all the activities using as few lecture halls as possible. Give an efficient greedy algorithm to determine which activity should use which lecture hall |         |    |   |   |   |   |           |    |    |    |   |   |

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| 22 |  <p>List applications of a minimum spanning tree. Find minimum spanning tree using Krushkal's algorithm of the following graph.</p> |
| 23 | Define graph. Describe strongly connected graph with example                                                                                                                                                         |
| 24 | Prove that if $G$ is an undirected bipartite graph with an odd number of vertices, then $G$ is non hamiltonian.                                                                                                      |
| 25 | <p>Find single source shortest path using Dijkstra's algorithm form a to e.</p>                                                   |
| 26 | <p>Find out the Minimum Spanning Tree using Kruskal Algorithm for given Graph</p>                                                 |
| 27 | <p>Find all pair of shortest path using Floyd's Algorithm for given graph</p>                                                     |

### UNIT-4 Exploring Graph

| SR NO. | Question Text                                                                                                                                                                                                                                                                         |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Explain with example how games can be formulated using graphs?                                                                                                                                                                                                                        |
| 2      | Write down the algorithm to determine articulation points in a given undirected graph. Give any application where it is applicable                                                                                                                                                    |
| 3      | Define: Acyclic Directed Graph, Articulation Point, Dense Graph, Preconditioning, Sparse Graph, Graph, Tree                                                                                                                                                                           |
| 4      | Explain Breadth First Traversal Method for Graph with algorithm.                                                                                                                                                                                                                      |
| 5      | Differentiate BFS and DFS.                                                                                                                                                                                                                                                            |
| 6      | How you can identify articulation points explain with example. Describe the use of articulation point.                                                                                                                                                                                |
| 7      | <p>Write an algorithm to find out the articulation points of an undirected graph.<br/>Find out articulation points for the following graph. Consider vertex A as the starting point.</p> <pre>graph TD; A --- B; A --- D; B --- C; C --- D; D --- E; D --- F; E --- F; C --- G;</pre> |
| 8      | Explain Depth First Traversal Method for Graph with algorithm.                                                                                                                                                                                                                        |
| 9      | Differentiate between depth first search and breadth first search.                                                                                                                                                                                                                    |
| 10     | Given an adjacency-list representation of a directed graph, how long does it take to compute the out-degree of every vertex? How long does it take to compute the in-degrees?                                                                                                         |

### **UNIT-5 Backtracking and Branch & Bound**

| <b>SR NO.</b> | <b>Question Text</b>                                                                                                                                                                                             |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | Explain Backtracking Method. What is N-Queens Problem? Give solution of 4-Queens Problem using Backtracking Method.                                                                                              |
| 2             | Explain Backtracking with Knapsack problem.                                                                                                                                                                      |
| 3             | Find an optimal solution to the knapsack instance $n=4$ , $M=8$ , $(P_1, P_2, P_3, P_4)=(3, 5, 6, 10)$ and $(W_1, W_2, W_3, W_4)=(2, 3, 4, 5)$ using backtracking. Also draw the corresponding state space tree. |
| 4             | Explain use of Branch & Bound Technique for solving Assignment Problem.                                                                                                                                          |
| 5             | For feasible sequence (7,5,3,1) solve 8 queen's problem using backtracking.                                                                                                                                      |
| 6             | Explain minimax principle with its use.                                                                                                                                                                          |
| 7             | Define feasible solution?                                                                                                                                                                                        |
| 8             | Prove that the fractional knapsack problem has the greedy-choice property.                                                                                                                                       |
| 9             | Define backtracking. State types of constraints used in backtracking.                                                                                                                                            |
| 10            | Differentiate branch and bound and back tracking algorithm                                                                                                                                                       |
| 11            | Draw the state space tree Diagram for 4 Queen problem                                                                                                                                                            |

### UNIT-6 String Matching

| SR NO. | Question Text                                                                                                                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | What are the basic ideas behind string matching? Which are the two popular algorithms for string matching? Explain and analyze any one in brief.                                                                                                  |
| 2      | What is the need of matching the string? Explain Finite automata for string matching. <u>OR</u> What is Finite Automata?Explain use of finite automata for string matching with suitable example.                                                 |
| 3      | Explain Rabin Karp method of string matching.                                                                                                                                                                                                     |
| 4      | Explain spurious hits in Rabin-Karp string matching algorithm with example. Working modulo $q=13$ , how many spuriou shits does the Rabin-Karp matcher encounter in the text $T = 2359023141526739921$ when looking for the pattern $P = 31415$ ? |
| 5      | Explain KMP string matching methods.                                                                                                                                                                                                              |
| 6      | Explain finite automata for string matching with example.                                                                                                                                                                                         |
| 7      | What is the basic idea behind Rabin – Karp algorithm? What is expected running time of this algorithm ? Explain it with example.                                                                                                                  |
| 8      | Give and explain Rabin-Carp string matching algorithm with example.                                                                                                                                                                               |
| 9      | Explain naïve string matching algorithm with example.                                                                                                                                                                                             |
| 10     | Show the comparisons the naive string matcher makes for the pattern $P=0001$ in the text $T=000010001010001$                                                                                                                                      |
| 11     | Working modulo $q=11$ , how many spurious hits does the Rabin-Karp matcher encounter in the text $T=3141592653589793$ when looking for the pattern $P=26$ ?                                                                                       |

### UNIT-7 Introduction to NP Completeness

| SR NO. | Question Text                                                                                                                         |
|--------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Define the class <b>P</b> and class <b>NP</b> . Is there any NP-Hard problem, which is also NP? If yes, give an example, if not, why? |
| 2      | Define :<br>(1) P (2) NP (3) NP-complete (4) NP-hard 5) Polynomial reduction                                                          |
| 3      | Write a brief note on NP-completeness and the classes-P, NP and NPC.                                                                  |
| 4      | Define P, NP, NP complete and NP-Hard problems. Give examples of each.                                                                |
| 5      | Explain Traveling salesman problem with example.                                                                                      |
| 6      | State whether travelling salesman problem is a NP-Complete problem? Justify your answer.                                              |

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